

STUDY GUIDE

CONCEPT SUMMARY

Big Ideas

- Arithmetic operations on polynomials are based on the arithmetic operations on whole numbers, and have similar properties.
- Multiplying and factoring are inverse processes, and a rectangle diagram can be used to represent them.

Applying the Big Ideas

This means that:

- Factors and multiples can be found for both whole numbers and polynomials.
- We multiply factors to determine their product.
- For trinomials, the factors can be constant terms, monomials, binomials, or trinomials. When we multiply the factors, we expand.
- We factor a polynomial by writing it as a product of its factors.
- Algebra tiles or a rectangle diagram can represent a product of polynomials.

Reflect on the Chapter

- Describe how the operations of addition, subtraction, and multiplication on whole numbers are similar to these operations on polynomials. Include examples in your description.
- Explain how a rectangle diagram is used to multiply and factor whole numbers and to multiply and factor polynomials. Why can the rectangle diagram be used for both multiplying and factoring?

SKILLS SUMMARY

Skill	Description	Example
Determine prime factors, greatest common factor (GCF), and least common multiple (LCM). [3.1]	Express a whole number as the product of its prime factors, using powers where possible. Use the prime factors to determine GCF and LCM.	As a product of prime factors: $64 = 2^6$ and $80 = 2^4 \cdot 5$ The GCF of 64 and 80 is: $2^4 = 16$ The LCM of 64 and 60 is: $2^6 \cdot 5 = 320$
Determine whether a number is a perfect square or a perfect cube. [3.2]	Use prime factors to check for perfect squares and perfect cubes.	$4225 = 5^2 \cdot 13^2$ Since the factors occur in pairs, 4225 is a perfect square and $\sqrt{4225}$ is: $5 \cdot 13 = 65$ 4225 is not a perfect cube because the factors do not occur in sets of 3.
Determine common factors for a polynomial. [3.3]	Look at the terms and determine their greatest common factor. Multiply the factors to verify.	$3x^2y - 21xy + 30y^2$ $= 3y(x^2 - 7x + 10y)$
Multiply binomials of the form $(x + a)(x + b)$ and $(ax + b)(cx + d)$. [3.4, 3.5, 3.6]	Use algebra tiles, diagrams, and the distributive property to multiply.	$(3d + 2)(4d - 5)$ $= 12d^2 - 15d + 8d - 10$ $= 12d^2 - 7d - 10$
Factor polynomials of the form $x^2 + bx + c$ and $ax^2 + bx + c$. [3.4, 3.5, 3.6]	Use algebra tiles, diagrams, and symbols to factor. Look for common factors first. Multiply the factors to verify.	$5x^2 - 9x - 2$ $= 5x^2 - 10x + x - 2$ $= 5x(x - 2) + 1(x - 2)$ $= (x - 2)(5x + 1)$
Multiply polynomials. [3.7]	Use the distributive property to multiply each term in the first polynomial by each term in the second polynomial.	$(x - 4)(3x^3 + 5x - 2)$ $= 3x^4 + 5x^2 - 2x - 12x^3 - 20x + 8$ $= 3x^4 - 12x^3 + 5x^2 - 22x + 8$
Factor special polynomials. [3.8]	Factor a perfect square trinomial and a difference of squares binomial. Look for common factors first. Multiply the factors to verify.	$25x^2 - 49y^2$ $= (5x + 7y)(5x - 7y)$ $4x^2 + 16xy + 16y^2$ $= 4(x + 2y)(x + 2y)$